OCNG 470 – Lab 1 Assignment

Introduction to Python

There are four parts: **1A–1D**. Complete all parts. Each part is worth **10 points**. Part 1B includes **+3 extra credit points**.
Submit on Canvas under Assignments for your section.

Lab Outcomes

You will submit:

1. One Python script file, `.py (or .ipynb)`
2. One PDF document

Required Data Files

Place the following files in your current Python working directory:

`R4901473_154.nc`
`Harvey_SST.txt`

The file `R4901473_154.nc` is the NetCDF file used for the Argo float temperature and pressure profile.
The file `Harvey_SST.txt` is used for the day-of-year and sea surface temperature data.

Submission Instructions

1. Complete the assignment in a new Python script file.

2. Save your Python script using the convention:

`Lab01_ZZ.py`

where:

- `01` = lab number
- `ZZ` = your initials

Example:

`Lab01_AB.py`

3. For questions that require a written answer, type your answers as comments in your Python script. In Python, comments begin with `#`.

4. Save your final submitted PDF using the naming convention:

`OCNG470_Lab01_Fall2026_SectionYYY_ZZ.pdf`

where:

- `YYY` = your section number
- `ZZ` = your initials

Example:

`OCNG470_Lab01_Fall2026_Section500_AB.pdf`

5. Upload the PDF to Canvas.

6. Before submitting, check that your Python script runs without errors from start to finish.

Python Packages Needed

This lab uses the following Python packages:

numpy
matplotlib
netCDF4

If you need to install them, you can use:

pip install numpy matplotlib netCDF4

or, if using Anaconda:

conda install numpy matplotlib netCDF4

Part 1A, 10 points

Getting Started with Python

In this section, you will create and run a simple Python script.

1. Open your preferred Python environment, such as Spyder, VS Code, JupyterLab, or a terminal.

2. Set your working directory to the folder where your Lab 1 files are saved.

3. Create a new Python script file.

4. Add the following section to your script:

```
# ---- Part 1A: Getting started with Python ----

# Name: Your Name
# Section: 500
# Lab Number: 1

name = "Your Name"
section = 500
lab_number = 1

print(f"Hello, my name is {name} from section {section}.")
```

5. Add a simple plot:

```
import numpy as np
import matplotlib.pyplot as plt

# Example variables
xxx = np.arange(1, 21)
yyy = xxx**2

plt.figure()
plt.plot(xxx, yyy, color="blue", linewidth=2, label="yyy")
plt.plot(xxx, yyy - xxx, color="red", linewidth=2, label="yyy - xxx")

plt.xlabel("x-axis [units]")
```

```
plt.ylabel("y-axis [units]")
plt.title("OCNG 470: Your Name: Lab 1A")
plt.legend()
plt.grid(True)

plt.savefig("Lab_1A.pdf")
plt.close()
```

This part introduces basic Python scripting, variables, comments, plotting, labels, legends, and PDF output.

Part 1B, 10 points + 3 extra credit

Working with NetCDF Files in Python

In this section, you will open and explore a NetCDF file.
The NetCDF file is:

R4901473_154.nc

In Python, you will use the `netCDF4` package.

1. Download the file `R4901473_154.nc` from Canvas.

2. Put the file in your current Python working directory.

3. Add a new section in your script labeled:
   ```
   # ---- Part 1B: Working with NetCDF files ----
   ```

4. Use the following code to open and inspect the NetCDF file:
   ```
   # ---- Part 1B: Working with NetCDF files ----

   from netCDF4 import Dataset

   nc_file = "R4901473_154.nc"

   # Open the NetCDF file
   nc = Dataset(nc_file, mode="r")

   # Display general information
   print(nc)

   # Display global attributes
   print("\nGlobal attributes:")
   for attr in nc.ncattrs():
       print(f"{attr}: {getattr(nc, attr)}")

   # Display dimensions
   print("\nDimensions:")
   for dim_name, dim in nc.dimensions.items():
       print(f"{dim_name}: size = {len(dim)}")
   ```

```
# Display variables
print("\nVariables:")
for var_name, var in nc.variables.items():
    print(f"{var_name}: dimensions = {var.dimensions}, shape = {var.shape}")
```

5. Answer the following questions as comments in your Python script.

```
# a. What is the title of this file?
# Answer:

# b. What is the long name for the attribute "JULD"?
# Answer:

# c. What are the units for latitude and longitude?
# Answer:

# d. What are the units for temperature?
# Answer:

# e. What are the units for pressure?
# Answer:

# f. Optional bonus: How many operational Argo floats are there today?
# Answer:

# g. Optional bonus: Where is this Argo float now?
# Answer:
```

6. Close the NetCDF file when done:

```
nc.close()
```

Part 1C, 10 points

Importing and Visualizing NetCDF Variables

In this section, you will read temperature and pressure from the NetCDF file and make a vertical profile plot.

1. Add a new section in your script labeled:

```
# ---- Part 1C: Importing and visualizing NetCDF variables ----
```

2. Read temperature and pressure from the NetCDF file:

```
# ---- Part 1C: Importing and visualizing NetCDF variables ----

from netCDF4 import Dataset
import numpy as np
import matplotlib.pyplot as plt

nc_file = "R4901473_154.nc"
```

```
# Open NetCDF file
nc = Dataset(nc_file, mode="r")

# Read temperature and pressure variables
temp = nc.variables["TEMP"][:]
pres = nc.variables["PRES"][:]

# Close NetCDF file
nc.close()
```

3. Convert the data to one-dimensional arrays and remove missing values:

```
# Convert to one-dimensional arrays
temp = np.array(temp[0]).flatten()
pres = np.array(pres[0]).flatten()

# Remove missing or invalid values
good = np.isfinite(temp) & np.isfinite(pres)

temp = temp[good]
pres = pres[good]
```

4. Make a plot with temperature on the x-axis and pressure on the y-axis.

Since this is a depth-profile-style plot, pressure should increase downward. Therefore, reverse the y-axis.

```
# Plot temperature versus pressure
plt.figure()
plt.plot(temp, pres, color="blue", linewidth=2)

plt.xlabel("Temperature (°C)")
plt.ylabel("Pressure (dbar)")
plt.title("Temperature vs Pressure")
plt.gca().invert_yaxis()
plt.grid(True)

plt.savefig("OCNG470_Lab01_Profile.pdf")
plt.close()
```

Part 1D, 10 points

Importing and Visualizing Text Data

In this section, you will read day-of-year and sea surface temperature data from a text file.
Use the filename:

Harvey_SST.txt

1. Add a new section in your script labeled:

```
# ---- Part 1D: Importing and visualizing text data ----
```

2. Read the text file:

```
# ---- Part 1D: Importing and visualizing text data ----

import numpy as np
import matplotlib.pyplot as plt

sst_file = "Harvey_SST.txt"

# Read two-column text file: DOY and SST
data = np.loadtxt(sst_file)

DOY = data[:, 0]
SST = data[:, 1]
```

3. Make a plot of the full dataset:

```
# Full SST plot
plt.figure()
plt.plot(DOY, SST, color="blue", linewidth=2)

plt.xlabel("Day of Year")
plt.ylabel("SST (°C)")
plt.title("Sea Surface Temperature: Full Record")
plt.grid(True)

plt.savefig("OCNG470_Lab01_SST_Full.pdf")
plt.close()
```

4. Make a zoomed-in plot using the following limits:

- x-axis: Day 190 to Day 280
- y-axis: 22°C to 32°C

```
# Zoomed SST plot
plt.figure()
plt.plot(DOY, SST, color="blue", linewidth=2)

plt.xlabel("Day of Year")
plt.ylabel("SST (°C)")
plt.title("SST: Zoomed, Day 190 to Day 280")
plt.axis([190, 280, 22, 32])
plt.grid(True)

plt.savefig("OCNG470_Lab01_SST_Zoom.pdf")
plt.close()
```